

# PHYSIVERSE

## Phase 2 – Information Architecture Document

Version: 1.0

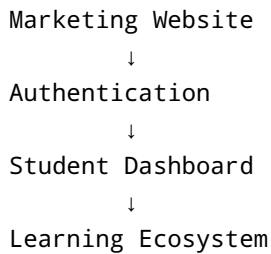
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### Objective

Design a scalable information architecture that allows students to navigate complex physics concepts through interactive experiences, simulations, and personalized learning paths.

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### Platform Architecture



```
graph TD; A[Marketing Website] --> B[Authentication]; B --> C[Student Dashboard]; C --> D[Learning Ecosystem];
```

Marketing Website  
↓  
Authentication  
↓  
Student Dashboard  
↓  
Learning Ecosystem

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### Primary Navigation

#### Public Navigation

Home

Explore Physics

Simulations

Virtual Labs

Universe Explorer

Community

Pricing

Blog

About

Contact

Login

Sign Up

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## Authenticated Navigation

Dashboard

Continue Learning

My Courses

My Simulations

Notes

Challenges

Achievements

AI Tutor

Settings

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## Complete Site Map

Home

|

|— About

|— Pricing

|— Blog

|— Contact

|

- └─ Learn
  - └─ Mechanics
  - └─ Thermodynamics
  - └─ Waves
  - └─ Electricity
  - └─ Magnetism
  - └─ Optics
  - └─ Modern Physics
  - └─ Astrophysics
- └─ Simulations
  - └─ Mechanics Lab
  - └─ Gravity Simulator
  - └─ Pendulum Simulator
  - └─ Wave Simulator
  - └─ Circuit Simulator
  - └─ Optics Simulator
  - └─ Space Simulator
- └─ Virtual Labs
- └─ Formula Explorer
- └─ AI Tutor
- └─ Universe Explorer
- └─ Challenges
- └─ Research Hub
- └─ Community
- └─ Dashboard

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## Student Journey

Landing Page

↓

Register

↓

Select Goal

↓

Diagnostic Assessment

↓

Personalized Roadmap

↓

Learn Concept

↓

Interactive Simulation

↓

Experiment

↓

Quiz

↓

Challenge

↓

Achievement

↓

Certificate

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## Learning Hierarchy

Subject

↓

Unit

↓

Chapter

↓

Topic

↓

Concept

↓

Simulation

↓

Practice

↓

Assessment

↓

Mastery

# Dashboard Architecture

Dashboard | | Overview | Continue Learning | Recommended Topics | Learning  
Statistics | XP & Streaks | Notes | Saved Lessons | Challenges | Achievements  
| Leaderboards | Settings

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## Teacher Dashboard

Teacher Dashboard | | Courses | Students | Assignments | Analytics | Virtual  
Labs | Reports | Certificates | Community

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## Homepage Information Architecture

### Section 1

Hero Universe

### Section 2

Physics in Real Life

### Section 3

Explore Physics Domains

### Section 4

Featured Simulations

### Section 5

Virtual Labs

### Section 6

Formula Explorer

## Section 7

AI Tutor

## Section 8

Community

## Section 9

Testimonials

## Section 10

Pricing

## Section 11

Footer

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# Learn Section Architecture

Learn | | Chapter List | Concept Page | Interactive Animation | Formula Breakdown  
| Simulation | Quiz | Notes | Challenge

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## Simulation Architecture

Simulation | | Instructions | Controls | Experiment | Observations | Explanation  
| Quiz

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## Formula Explorer Architecture

Formula | | Formula Visualization | Variables | Interactive Controls | Graph |  
Examples | Real World Applications

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## AI Tutor Architecture

AI Tutor | |—— Ask Question |—— Explain Concept |—— Generate Quiz |—— Solve Problems |——  
Voice Mode |—— Study Recommendations

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## Community Architecture

Community | |—— Discussions |—— Study Groups |—— Shared Notes |—— Challenges |——  
Leaderboards

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## Mobile Navigation

Home

Learn

Labs

Explore

Profile

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## Search Architecture

Global Search

Search:

- Concepts
  - Formulas
  - Simulations
  - Articles
  - Notes
  - Labs
-

# Future Architecture

AR Learning

VR Learning

School Portal

Research Portal

Teacher Marketplace

Content Marketplace

Physics Competition Platform

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## Architecture Principles

1. Learn by Seeing
2. Learn by Doing
3. Learn by Experimenting
4. Learn by Exploring
5. Learn by Teaching
6. Learn by Playing